RAVI SRIVASTAVA

Curriculum Vitae

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Education

2024–2026 : **Master of Technology, Data Science & Artificial Intelligence**, *Indian Institute of Information Technology*, Ranchi,India.

CGPA : 8.79/10

2018–2022 : **Bachelor of Technology, Information Technology**, *North-Eastern Hill University*, Shillong,India.

CGPA : 7.85/10

Publications

Journal Articles

July 2025 **Ravi Prakash Srivastava**, Sanjay Dahiya, and Aashish Nehra. Iot-hits: An iot based human intrusion detection system for border region using deep learning. ***IETE Journal of Research***, volume 71. Taylor & Francis, July 2025.

March 2022 Ajay Kumar, Mohammad Amir Khusru Akhtar, Abhishek Pandey, and **Ravi Prakash Srivastava**. Smart city vehicle accident monitoring and detection system using (mems, gsm, gps) raspberry pi 4. ***IETE Journal of Research***, volume 69, pages 8121–8129. Taylor & Francis, March 2022.

Preprints

July 2025 Jasmaine Khale and **Ravi Prakash Srivastava**. Balanced few-shot episodic learning for accurate retinal disease diagnosis, July 2025. Preprint, arXiv:2512.04967v1.

In Conference Proceedings

2026 **Ravi Prakash Srivastava**, Sourabh Sah, Kirti Kumari, Pavan Vovveti, and Md. Shah Fahad. Adversarial prompt injection attacks on large language models: Cryptographic key leakage and defense strategies. In *Proceedings of ICACA 2026*, Ranchi, Jharkhand, India, March 2026. Springer Nature, Lecture Notes in Networks and System [In press].

2026 Sourabh Sah, **Ravi Prakash Srivastava**, Kirti Kumari, and Md. Shah Fahad. Deepfake video detection using face-centric processing and frame sampling. In *Proceedings of the International Conference on Advances in Computing and Applications (ICACA 2026)*, Ranchi, Jharkhand, India, March 2026. Springer Nature, Lecture Notes in Networks and System [In press].

Research Experience

March 2026 – Present **Software Engineer**, *SS Software Solutions LLC, Hyderabad, India.*

- Designed and standardized ML pipeline infrastructure with model versioning (**MLflow/DVC**), JSON schema-based structured logging, observability monitoring, and drift detection, improving reliability across production ML models and reducing incident diagnosis time by ~40%.
- Integrated security and governance practices into ML workflows by implementing **STRIDE-based threat modeling**, SAST tools (**Bandit, Semgrep**), and dependency vulnerability scanning within CI/CD pipelines (**GitHub Actions**), reducing security review effort and improving deployment reliability.

August 2022 – February 2023 **Machine Learning Intern**, *C-DAC (Government of India), Hyderabad, India.*

- Developed machine learning models for predicting student academic performance using educational data analytics.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.

Research Projects

2026 **Cross-Domain Text Summarization Using Transformer Models**, *Master's Thesis.*

- Investigated domain shift and catastrophic forgetting by adapting **BART-large-CNN** from news summarization to legal documents using the **BillSum** dataset.
- Implemented parameter-efficient fine-tuning (**LoRA**) with **Elastic Weight Consolidation (EWC)** to preserve pre-trained knowledge during domain adaptation.
- Achieved improved cross-domain generalization while fine-tuning only **1–2%** (~2M of 406M) of model parameters, significantly reducing GPU memory usage and training cost.

2025 **LLM Semantic Cache & Analytics Dashboard**, *Live Demo.*

- Designed and implemented an embedding-based semantic caching layer for Large Language Models to retrieve responses using semantic similarity instead of exact keyword matching.
- Reduced LLM API latency and inference costs through intelligent cache reuse while maintaining response quality.

Fellowships & Awards

M.Tech Recipient of the ***Institution of Electronics and Telecommunication Engineers (IETE) Scholarship*** for academic excellence during 2-year M.Tech. studies in Computer Science and Engineering.

2026 ***Best Paper Award*** at ICACA 2026 for the paper, "Adversarial Prompt Injection Attacks on Large Language Models: Cryptographic Key Leakage and Defense Strategies."

Academic Achievements & Recognitions

2026 Served as a **Reviewer** for the **International Conference on Innovations in Computing, Communication, and Sustainable Technologies (ICICST 2026)**, organized by the Department of Information Technology, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, held April 3–5, 2026.

2022 Awarded **Associate Membership (AMIETE)** of the **Institution of Electronics and Telecommunication Engineers (IETE), India**.

Technical Skills

Programming Languages: Python, C++, Java, JavaScript

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, Image Processing, Deepfake Detection, Large Language Models (LLMs), Hugging Face Transformers, PyTorch, Scikit-learn, OpenCV, Keras

MLOps & DevOps: Git, Docker, GitHub Actions (CI/CD), MLflow, DVC

Databases: SQL, MySQL, MariaDB, Relational Database Design

Language Skills

English,Hindi